

Sphatik OS — Design Specification

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Vision and design principles

Sphatik (स्फटिक, Sanskrit for crystal) is a phone OS where content is the hero and the interface is a layer of living glass that floats above it, bends light, and gets out of the way. It takes the freedom of Android and the polish of iOS, then adds an AI-native Intent Bar as its own signature.

One-line pitch: open like Android, fluid like iOS, and it understands what you mean.

1. **Content first, chrome second.** UI sits on glass layers above content, never on opaque bars. Controls shrink or fade while you scroll or watch.
2. **Physics, not animation curves.** Every movement is a spring with mass and velocity. Anything you can see moving, you can grab and interrupt mid-motion.
3. **One gesture language everywhere.** The same swipe means the same thing in every app and system surface.
4. **Open by default.** Real file system, default-app choice, sideloading, widgets anywhere, and no lock-in to one store.
5. **Private by design.** Permissions are granted per use, visible in the status area, and reviewed in one dashboard.
6. **Say it, don't hunt for it.** The Intent Bar lets you type or speak a goal and the OS composes the answer across apps.
7. **Calm.** No badges screaming for attention. Notifications are grouped, summarized, and scheduled.
8. **Fast on modest hardware.** 60 fps locked on the Redmi Note 9 Pro (120 fps on high-refresh screens later), glass effects degrade gracefully instead of stuttering.

Signature details that make Sphatik its own (not a copy):

- **Melt glass:** when two glass panels touch, their edges merge like liquid droplets, then split apart when pulled away.
- **Caustics:** moving glass casts a faint, shifting light pattern on the content beneath it.
- **Gyro light:** specular highlights on glass edges follow the tilt of the phone.
- **Halo:** a live-activity ring that flows around the camera cutout instead of a pill shape.

Design language: the glass material system

All system UI is built from four glass materials stacked on a Z-axis, so depth tells you what is on top and what you can touch.

Material	Used for	Blur radius	Tint opacity	Refraction	Z-level
Clear	Buttons over media, Halo, floating toolbars	8 px	5%	Strong edge bend	3
Regular	Tab bars, widgets, notification cards, dock	24 px	18%	Medium	2
Thick	Control Center, sheets, keyboard, menus	40 px	35%	Light	4
Frosted	Lock screen overlays, dimmed backdrops behind modals	64 px	55%	None	1

How one glass panel is rendered (per frame, in the compositor):

1. **Sample** the pixels behind the panel from the previous layer.
2. **Blur** with a dual-Kawase pass at the material's radius.
3. **Refract** near the edges: a signed-distance field of the panel shape bends the background inward by up to 6 px, so rounded rims look like thick glass.
4. **Tint** with the adaptive color (see Color section), stronger when content behind is busy or bright.
5. **Light**: a 1 px inner rim highlight plus a soft specular spot whose position follows the gyroscope.
6. **Disperse**: a 0.5 px red/blue split on the outermost rim only, for a hint of prism.
7. **Shadow**: a soft ambient shadow, larger for higher Z-levels.

Shape rules

- Corners use continuous curvature (squircle), never plain circular arcs.
- Corner radius of any panel = display corner radius minus its inset from the screen edge, so inner shapes stay concentric with the device.
- Standard radii: 12 (chips), 20 (buttons, small widgets), 28 (cards, large widgets), 36 (sheets).

Behavior rules

- **Legibility guard**: if text contrast on glass drops below 4.5:1, tint opacity increases automatically until it passes.
- **Scroll response**: tab bars and toolbars shrink to a compact pill after 40 px of downward scroll and expand on upward scroll.

- **Melt:** panels closer than 8 px merge with a metaball blend; pulling them apart stretches then snaps the bridge.
- **Reduce Transparency** setting swaps every material for a solid color at the same tint. Low Power mode drops refraction and caustics first, then halves blur radius. Where true refraction is not available (browsers, very weak GPUs), glass falls back to a lens-light treatment: a bright top hairline, soft inner glow, bottom caustic, diagonal mirror sheen and a specular spot that follows tilt.

Color, typography, iconography

Color comes from your wallpaper, type comes from one variable font, and icons share one grid so the whole system feels made by one hand.

Color

- **Adaptive palette:** on wallpaper change, the OS extracts 1 primary, 1 secondary and 1 neutral hue, then builds 13-step tonal ramps for each (like Android’s dynamic color, but applied to glass tint too).
- **Semantic tokens** (apps use these, never raw hex): `surface` , `on-surface` , `accent` , `on-accent` , `success` , `warning` , `danger` , `glass-tint` , `glass-rim` .
- **Light, Dark, and Auto** modes; Auto follows sunset or a schedule.
- **Icon tinting options:** Original, Tinted (monochrome in accent), Clear (glass icons).

Token	Light mode	Dark mode
surface	Neutral 98	Neutral 8
on-surface	Neutral 10	Neutral 94
accent	Primary 45	Primary 75
glass-tint	White at material opacity	Neutral 12 at material opacity
glass-rim	White 70%	White 22%

Typography

- One custom variable font family for the system (see the Sphatik Sans section), **Sphatik Sans**, with weight 100 to 900 and an optical-size axis, plus **Sphatik Mono** for code and numbers in the Calculator.
- Supports Latin, Devanagari and other Indic scripts from day one.

Style	Size / line height (pt)	Weight	Use
Display	96 / 96	200 to 700 (user picks)	Lock screen clock
Title 1	34 / 41	700	App page titles
Title 2	22 / 28	600	Section titles, sheet titles
Headline	17 / 22	600	Card titles, list headers
Body	17 / 22	400	Default text
Callout	15 / 20	400	Secondary text
Caption	12 / 16	500	Labels under icons, timestamps

All sizes scale with the system Text Size setting (80% to 310%).

Iconography

- **App icons:** 1024 px master on a 24-column grid, squircle mask applied by the system, with 3 layers (background, midground, foreground) so the OS can add parallax and glass depth.
- **System symbols:** a line-icon set drawn on a 24 px grid with a 1.75 px stroke, in 3 weights matching text weight.
- **Rule:** an icon never carries text inside it, except the Calendar date.

Sphatik Sans: the custom typeface

Sphatik Sans is a minimal, low-contrast sans with one quiet signature: tiny facet cuts, like the edges of a cut crystal, that you notice only at large sizes.

Concept: "cut crystal"

- **Skeleton:** geometric-humanist. Round letters (o, c, e) are near-circular for calm and elegance; open apertures keep it readable at small sizes.
- **Facet terminals:** stroke ends are cut at a gentle 8° angle instead of flat, like light catching a crystal edge. At text sizes the cut softens to almost flat.
- **Diamond dots:** the dots on i, j, the period, colon and exclamation mark are small rhombuses at display sizes and round at text sizes. This is the font's signature, and it echoes the OS name.

- **Contrast:** almost none in Text (about 5%), rising to about 18% in Display, where thin joins make the clock and titles feel refined.
- **No decoration beyond that.** No spurs, no ink traps visible to the eye, no quirky alternates in the default set.

Signature glyphs

Glyph	Default shape	Why
a	Double-story in Text, single-story in Display	Legible small, elegant large
g	Single-story, open tail	Clean and minimal
t	Top cut with an 8° facet	Carries the crystal cut into running text
y	Straight diagonal tail	Minimal, no curl
R	Straight leg from the bowl	Crisp, architectural
Q	Short detached diagonal tail	Light and precise
1	Flag, no base serif	Clean on the lock screen clock
0	Oval; slashed variant in Mono	Readable digits, clear code
₹	Drawn to match stroke and facet style	First-class Indian currency sign

Metrics (units per em 1000)

Metric	Value
Cap height	700
x-height	520 (tall, for small-size legibility)
Ascender / descender	760 / -240
Default tracking, Text sizes	+1%
Default tracking, Display sizes	-2.5%

Variable axes

- **Weight (wght):** 100 to 900, so the lock screen clock can go ultra-thin.
- **Optical size (opsz):** 12 to 96. The OS sets it automatically from point size, switching dots, contrast, spacing and the a shape.

- **Grade (GRAD):** -50 to +50. Makes letters slightly bolder without changing width, applied automatically when text sits on Clear glass or in Dark mode, so glass never shifts the layout.

Family

1. **Sphatik Sans:** Latin, Latin Extended, Greek basics, arrows, math, and ₹.
2. **Sphatik Devanagari:** designed alongside the Latin, with the shirorekha (headline) weight matched to Latin stems and the same facet cut on terminals, so Hindi and English look like one family in mixed text.
3. **Sphatik Mono:** same skeleton at a fixed 600-unit width, slashed zero, for code, the Calculator and timers.
4. **Tabular figures** are on by default for clocks, timers and battery %, so digits don't jitter as they change.

How to build it

1. **Sketch control glyphs** first: n, o, H, O, a, g, e, t, and the diamond dot. These set the whole character.
2. **Draw 6 masters** in Glyphs 3, or free tools like Fontra or FontForge: Thin, Regular and Black, each at Text and Display optical sizes.
3. **Expand** to basic Latin plus figures and punctuation (about 250 glyphs), test in the OS prototype, then extend to the full set (about 900 glyphs).
4. **Devanagari:** draw with a Devanagari type designer, or start from an OFL-licensed base such as Noto Sans Devanagari and adapt its terminals and headline to match.
5. **Faster start option:** fork an OFL font like Inter or Manrope, then redraw the terminals, dots and signature glyphs. OFL allows this, but the result must stay under OFL and use a different name.
6. **Release** the finished family under the SIL Open Font License, so app developers can use it freely.

Realistic effort: about 2 to 3 months part-time for the Latin family, and 3 to 6 more months for Devanagari.

Wallpapers and boot identity

Four built-in wallpapers, all generated as vector art so they stay sharp at any resolution and take almost no storage.

Wallpaper	Look	Notes
Liquid Aurora (default)	Silky ribbons of violet, cyan, rose-gold and ice-blue on deep ink	Satin shading, a mirror highlight along each ribbon edge, soft glow and fine grain; the top half stays dark so the clock reads clearly
Obsidian	The same flowing ribbons, mirrored, in black and polished graphite	Faint cyan and lilac reflections; the most minimal option
Quartz Dawn	Cut crystal lit from inside on a dusk gradient	Turns on the depth effect: the crystal tip passes in front of the lock-screen clock
Amethyst	The crystal on a violet night sky	Depth effect as above

Boot screen: black, with the SKY wordmark settling out of a blur while a band of crystal light sweeps across the letters, centred exactly in the middle of the screen, and "SPHATIK OS" sits quietly at the bottom. It then fades into the lock screen. Holding the side button restarts.

Motion system and global gestures

Every transition is a spring driven by your finger's velocity, and every gesture below works the same in every app.

Spring presets (damping ratio / response time)

Preset	Damping	Response	Used for
Snappy	0.90	0.25 s	Buttons, toggles, small chips
Smooth	1.00	0.40 s	Sheets, cards, page pushes
Bouncy	0.72	0.45 s	App open/close, Halo expansion, Melt
Gentle	1.00	0.60 s	Wallpaper parallax, lock-to-home fade

Motion rules

- **Continuity:** an app opens by growing out of the exact icon, widget, or notification you tapped, and closes back into it.
- **Interruptible:** any animation can be caught mid-flight and reversed with no jump.

- **Rubber-banding:** scrolling past an edge resists at 55% of finger travel and springs back.
- **Haptics:** light tick on toggles and snapping, medium on long-press menus, heavy only for errors and destructive confirms.
- **Reduce Motion** replaces scale and zoom transitions with cross-fades and turns off parallax, caustics and Melt.

Global gestures (full-screen gesture navigation; a 3-button bar is available in Settings for accessibility)

Gesture	Where	Result
Swipe up from bottom edge	Anywhere	Go Home
Swipe up and hold	Anywhere	App Switcher
Swipe left/right along bottom edge	In an app	Jump to previous/next app
Swipe inward from left or right edge	In an app	Back (Android-style, works on both sides)
Swipe down from top-left	Anywhere	Notification Center
Swipe down from top-right	Anywhere	Control Center
Swipe down on home screen middle	Home	Intent Bar
Long-press any icon, widget, card	Anywhere	Context menu with glass blur
Long-press empty home space	Home	Edit mode
Double-tap status bar	In an app	Scroll to top
Three-finger swipe left/right	In an app	Undo / Redo
Tap and hold Halo	Anywhere	Expand live activity
Double-tap side button	Anywhere (configurable)	Quick launch camera, wallet, or torch

Edge gestures have a 20 px trigger zone. Apps that need edge swipes (drawing, games) can request a one-time "confirm" mode where a second swipe is needed.

Global layout system: grid, safe areas, sizes

All layouts are built on a 4 pt base unit and a 4-column phone grid, measured in density-independent points so one design fits every screen.

Reference canvas: 393 x 852 pt (a typical 6.1" phone at 3x density). Designs scale by width; heights reflow.

Spacing scale (pt): 4, 8, 12, 16, 20, 24, 32, 40, 56, 72.

Grid

- Phone portrait: 4 columns, 16 pt side margins, 12 pt gutters.
- Phone landscape and foldables (unfolded): 8 columns, 24 pt margins, 16 pt gutters.
- Tablets later: 12 columns.

Safe areas

Zone	Height	Rule
Status area (with Halo)	54 pt	Only system indicators and Halo; apps draw content beneath it, never controls
Gesture area	34 pt	Bottom strip; no tappable controls, only the home handle
Side edge zones	20 pt each	Reserved for Back swipe
One-hand reach zone	Bottom 60% of screen	Primary actions of every system surface must live here

Sizing rules

- Minimum touch target 44 x 44 pt, even when the visible element is smaller.
- Minimum 8 pt between two touch targets.
- Line length for body text: 40 to 75 characters; wider screens add margins, not longer lines.

Standard app structure (every app follows this unless it is full-screen media or a game)

1. **Top:** a large title that collapses into a compact glass title bar on scroll.
2. **Middle:** content, scrolling edge-to-edge beneath all glass.
3. **Bottom:** a floating glass tab bar (2 to 5 tabs) with a separate floating search or action button to its right. On scroll down it shrinks to a pill showing only the active tab.

Putting navigation at the bottom, not the top, is a deliberate rule: it keeps every primary action in thumb reach on tall phones.

Size classes so apps adapt: Compact (phone portrait), Medium (phone landscape, small foldable), Expanded (unfolded foldable, split-screen at 50%+).

Complete feature set, v1 vs later

v1 is what a person needs to use Sphatik as their only phone; everything else waits until that works.

Area	Feature	Inspired by	Release
System UI	Glass material system, Melt, gyro light, caustics	Own	v1
System UI	Adaptive wallpaper color for tint, accents, icons	Android	v1
System UI	Lock screen with editable clock, 4 widget slots, depth effect	iOS	v1
System UI	Free-placement home screen, widgets anywhere	Android	v1
System UI	App Library with auto categories and search	iOS	v1
System UI	Halo live activities around camera	Own (idea from iOS)	v1
System UI	Split Notification Center and Control Center	Both	v1
System UI	Customizable Control Center tiles	Both	v1
Navigation	Gesture nav with Back from both edges + 3-button option	Android	v1
Multitasking	Card app switcher, quick app flip along bottom edge	iOS	v1
Multitasking	Split screen (2 apps) and floating mini-window	Android	v2
AI	Intent Bar: search apps, files, contacts, settings, web	Own	v1
AI	Intent Bar actions across apps ("send my ETA to Mom")	Own	v2

Area	Feature	Inspired by	Release
AI	Notification summaries and smart grouping	Both	v2
AI	On-device model, cloud only with explicit opt-in	Own	v2
Notifications	Channels per app, snooze, scheduled summary	Android	v1
Notifications	Focus modes that filter people, apps, and home pages	iOS	v2
Freedom	Real file manager with folders	Android	v1
Freedom	Choose default browser, launcher, keyboard, dialer, SMS	Android	v1
Freedom	Sideload apps, multiple app stores	Android	v1
Privacy	Per-use permissions, one-time location, approximate location	Both	v1
Privacy	Camera/mic/location indicators in status area	iOS	v1
Privacy	Privacy dashboard: 7-day access log per app	Android	v1
Privacy	App sandboxing with per-app network toggle	Own	v2
Phone basics	Calls, SMS, contacts, dual SIM, VoLTE	Standard	v1
Phone basics	Camera with photo, video, portrait later	Standard	v1 (portrait v2)
Phone basics	Wi-Fi, Bluetooth, hotspot, GPS, NFC	Standard	v1 (NFC payments v3)
Keyboard	Swipe typing, clipboard history, Hinglish and Hindi	Android	v1
Continuity	Clipboard and file sharing with a desktop app	iOS	v3
Security	Encrypted storage, verified boot, face/fingerprint unlock	Both	v1 (face v2)
Security	Separate Work profile	Android	v3

Area	Feature	Inspired by	Release
Updates	A/B seamless updates with rollback	Android	v1
Accessibility	Screen reader, text scaling, voice control, color filters	Both	v1 (voice control v2)
Developer	Sphatik SDK, UI toolkit, app packaging, store	Own	v1 alpha
Compatibility	Run Android apps in a container	Waydroid-style	v2

Android app compatibility is the single biggest factor for real-world adoption; plan for it early even if it ships in v2.

Lock screen layout and behavior

The lock screen is a personal poster: wallpaper, a large glass clock, up to 4 widgets, and notifications that stay out of the way until you want them.

Layout (top to bottom)

1. **Status area (54 pt):** carrier/time left, Halo center, signal/battery right.
2. **Date line (20 pt):** "Sun, 27 Sep" plus one optional inline widget (weather, next event, or battery).
3. **Clock (96 to 160 pt tall):** Display font, user-chosen weight and color, rendered in Clear glass. With Depth Effect on, the wallpaper's subject (detected by an on-device segmentation model) overlaps the clock.
4. **Widget row (72 pt):** three minimal glass complications: a round weather dial, a round battery ring and a slim capsule for the next event. Each can be switched off.
5. **Flexible space:** wallpaper shows through.
6. **Notification stack:** bottom-anchored, collapsed to a count ("5 notifications") by default; swipe up to expand into a list.
7. **Quick action row (56 pt above gesture area):** 2 round Clear-glass buttons, bottom-left and bottom-right, default Torch and Camera, both replaceable.
8. **Home handle** in the gesture area.

States

State	What shows	Transition in
Always-on (screen off)	Clock outline, date, Halo, notification icons; wallpaper dimmed 90%; 1 Hz refresh	Screen timeout
Wake (locked)	Full layout, notification content hidden	Raise, tap, or button; Gentle spring fade-up
Unlocked, not dismissed	Notification content revealed, padlock icon opens	Face/fingerprint match
Unlocking to home	Clock and widgets lift and fade, home icons fly in from depth	Swipe up; Bouncy spring

Interactions

- Swipe up: unlock (or show PIN pad on Thick glass if biometrics fail).
- Swipe left: open Camera directly (works without unlocking, gallery limited to this session's shots).
- Swipe right: Today view (a stack of widgets).
- Long-press anywhere: customise mode. The lock screen shrinks, the clock and widgets get dashed outlines, and a glass sheet slides up with the options below; swipe sideways to switch between saved lock screens, each linkable to a Focus mode.
- Press and hold a quick action button: triggers with a haptic, prevents pocket presses.

Customization options: three clock styles (Sphatik, Serif, Rounded), a weight slider from 100 to 800, six colours, Solid or Glass finish (Glass draws the digits as a hollow, lit outline), widget toggles and wallpaper. Choices persist across restarts.

Home screen and widgets

Home is a grid you can place anything on freely, with a floating glass dock and pages you swipe between, ending in the App Library.

Layout

1. **Status area** (54 pt).
2. **Icon grid:** 4 columns x 6 rows on the reference canvas (5 x 6 optional density). Icons 60 pt, labels in Caption below, labels hideable.
3. **Page dots** (16 pt) that turn into a scrubber when you drag across them.
4. **Dock:** a floating Regular-glass bar, 16 pt from the sides, 4 to 5 icons, 88 pt tall, radius concentric with the display.

5. Gesture area.

Placement rules

- Icons and widgets can sit in **any cell**, including leaving gaps (Android freedom), with an optional "auto-arrange" toggle that packs them from the top (iOS tidiness).
- Folders: drop one icon on another; folder opens as a Thick-glass panel with a 3 x 3 grid per page.
- Pages: up to 12. The last page to the right is always the App Library.
- Swipe right from the first page: Today view (a widget stack).

Widget sizes (grid cells)

Size	Cells	Typical content
Small	2 x 2	One glanceable value: weather, battery, step count
Medium	4 x 2	A short list: next 3 events, 4 contacts
Large	4 x 4	Rich view: month calendar, photos, music player
Wide strip	4 x 1	Search bar, quick toggles, a ticker

Widget rules

- Widgets are **interactive**: buttons and toggles work in place without opening the app.
- Smart Stack: drop widgets on each other to stack; the stack rotates to the most relevant one by time and usage.
- Widgets render on Regular glass by default; apps can opt into Clear for photo widgets.
- Refresh budget: widgets update at most every 15 minutes unless showing a live activity.

Edit mode (long-press empty space)

- Icons wobble gently and show a remove badge; widgets show resize handles on corners.
- An edit bar replaces the search pill with Add widget and Done. Add widget opens a gallery sheet with live previews of Weather, Up next, Battery, Now playing and Clock; the top row holds two square widgets. Removing an app from Home keeps it in the App Library.
- Dragging an icon to the screen edge flips pages after a 400 ms hover.
- Pages view: pinch in to see all pages as thumbnails and reorder, hide, or delete them.

App Library, search and Intent Bar

The App Library holds every installed app automatically, and the Intent Bar is one box that finds anything and, later, does things for you.

App Library layout

1. A search field in Regular glass pinned at the **bottom** (thumb reach), not the top.
2. Above it, auto-generated category tiles in a 2-column grid: Suggestions, Recently Added, Social, Productivity, Entertainment, Utilities, and so on. Each tile shows 3 large icons plus a cluster of 4 small ones.
3. Tap a big icon to launch; tap the cluster to open the whole category.
4. Swipe down or tap search: switches to an A to Z list with a letter scrubber on the right edge.

Intent Bar (swipe down on home, or tap the App Library search)

The bar opens as a Clear-glass pill that Melts out of the status area and stretches to full width at the top of the keyboard.

Mode	You type or say	Sphatik returns	Release
Find	"wifi", "rahul", "invoice pdf"	Grouped results: Apps, Contacts, Settings, Files, Messages, Web	v1
Quick answer	"45 x 18", "100 usd in inr", "weather tomorrow"	An inline answer card; no app launched	v1
Action	"send my ETA to Mom", "turn on hotspot and share password with Aman"	A preview card listing the steps and the apps it will use, with one Confirm button	v2
Ask	"what did Priya say about Friday?"	An answer drawn from your messages and mail, citing the source messages	v2

Intent Bar rules

- Results stream in as you type, fastest sources first (apps, settings) then slower ones (files, web).
- Every Action shows exactly what will happen and needs one confirming tap; nothing that sends, pays, or deletes happens silently.
- Apps expose actions to the Intent Bar through the SDK by declaring **intents** (e.g. `message.send` , `ride.book`) with typed parameters.

- Personal data stays on-device; a web or cloud model is used only when you enable it, and results from it are labeled.
- A mic button on the right of the bar switches to voice; results appear the same way.

The Halo: live activities around the camera

The Halo turns the camera cutout into a live status ring that grows into a glass panel when something is happening, so ongoing tasks never need a notification.

States

State	Size	Looks like	Example
Idle	Cutout only	Nothing drawn	Nothing running
Ring	Cutout + 3 pt ring	A thin glowing ring whose color and fill show progress	Download at 60%, timer counting
Compact	36 pt tall capsule	Ring plus a small icon left and short text right	Call 02:14, song playing, ride 4 min away
Split	Compact + a small detached bubble	A second activity Melts off as its own bubble	Timer and music at the same time
Expanded	Up to 393 x 200 pt panel	Clear-glass panel with full controls	Media controls, live cricket score, delivery map

Behavior

- Tap: opens the owning app. Long-press or swipe down on it: Expanded. Swipe up or tap outside: back to Compact.
- Swipe it sideways: swaps which activity is primary when two are running.
- Alerts (incoming call, Face unlock, charging, AirDrop-style share) Bounce out to Expanded briefly, then settle back.
- Priority order when many activities run: call > navigation > timer > media > downloads. Built so far: calls show a green phone and running duration with Mute and End; timers show an amber progress ring with Pause and Cancel; media shows album art and an animated equaliser.
- Maximum 2 visible activities; others collect in Notification Center.

For devices without a hole-punch camera, the Halo sits at top-center as a floating Clear-glass capsule with the same states.

For app developers: apps start a Live Activity through the SDK with 4 templates (Compact text, Progress, Media, Custom view). Updates are pushed at most once per second; the system owns the animations.

Notifications

Notifications arrive as small glass banners, collect in a grouped center you pull from the top-left, and can be batched so your phone stays calm.

Banner (while using the phone)

- Drops from the Halo as a Regular-glass card, 16 pt side margins, max 3 lines.
- Auto-dismiss after 4 s. Swipe up to dismiss, swipe down to expand with reply field and action buttons, tap to open.
- During full-screen video or games, only calls and alarms show as banners; everything else waits.

Notification Center layout (swipe down from top-left)

1. The backdrop becomes Frosted glass over the current screen.
2. Header row: date, and a "Clear all" button that needs a second tap to confirm.
3. **Live section:** ongoing items (music, downloads, navigation) pinned at top.
4. **People section:** messages and calls, grouped by person, avatar on each group.
5. **Apps section:** grouped by app, collapsed to the latest item with a count.
6. **Summary card** (if scheduled): the batched low-priority items.

Card interactions

Gesture	Result
Tap	Open the app at that item
Swipe right (short)	Mark as read / dismiss
Swipe left (short)	Reveal Snooze (15 min, 1 h, tonight, tomorrow) and Settings
Swipe left (long)	Dismiss whole group
Long-press	Channel settings: mute this type, deliver quietly, turn off
Tap group	Expand the group into individual cards

Importance levels apps pick per channel, and users can override:

1. **Urgent:** banner, sound, can break through Focus (calls from favorites, alarms).
2. **Normal:** banner and sound.
3. **Quiet:** goes straight to Notification Center, no banner, no sound.
4. **Summary:** held and delivered in the scheduled summary (default 9 am and 7 pm).

Apps never show red badge counts on icons by default; a small accent dot is optional per app.

Control Center

Control Center drops from the top-right as a single Thick-glass sheet of tiles you can rearrange, resize, and spread across pages.

Layout (top to bottom, 4-column tile grid, 12 pt gaps)

1. **Header:** battery %, network name, an Edit (pencil) button, and a Settings shortcut.
2. **Connectivity block (2 x 2):** Wi-Fi, Mobile data, Bluetooth, Airplane. Tap toggles; long-press expands in place into a list of networks or devices.
3. **Media tile (4 x 2):** artwork, title, scrubber, play controls, output device picker.
4. **Sliders (1 x 3 each, vertical):** Brightness and Volume. Drag anywhere on the slider; the glass fills like liquid rising.
5. **Quick tiles (1 x 1):** Torch, Rotation lock, Do Not Disturb/Focus, Hotspot, Screen record, Battery saver, Dark mode, Camera, Calculator, QR scanner.
6. **Page dots:** swipe left for a second page (Smart home, extra tiles).

Tile sizes: 1 x 1, 2 x 1, 2 x 2, 4 x 2, and 1 x 3 (vertical slider). A tile that grows shows more controls, e.g. Torch at 2 x 1 gets a brightness slider.

Behavior

- The sheet follows your finger as you pull; release past 30% of the screen height and it springs open, less and it springs back.
- Pulling Control Center over Notification Center (or vice versa) switches between them with a Melt transition.
- Tiles that are on use the accent color filling the glass; off tiles stay clear.
- Long-press any tile: its expanded view on Thick glass, or deep settings.
- Edit mode: tiles wobble; drag to reorder, pull a corner to resize, tap + to add from all available tiles, including third-party app tiles.

Multitasking and app switcher

You flip between apps with one swipe along the bottom, see them all as floating glass cards, and in v2 run two at once.

App Switcher layout (swipe up and hold)

1. The current app shrinks to a card at 70% scale, centered, with its icon and name above it.
2. Other recent apps line up horizontally, overlapping slightly, most recent on the right.
3. Background: Frosted glass over the wallpaper.
4. Bottom row: "Clear all" and a Split button (v2).

Switcher interactions

Gesture	Result
Swipe sideways	Scroll through apps
Tap a card	Open that app (card grows back to full screen)
Flick a card up	Close the app
Long-press a card	Menu: App info, Pin, Split with..., Floating window
Drag a card down	Pin it to the left of the list so it stays first

Quick flip: swipe left or right along the bottom edge to move between recent apps without opening the switcher. Stop mid-swipe and the two apps sit side by side until you let go.

Split screen (v2)

- Two apps stacked top/bottom in portrait, left/right in landscape.
- A glass divider handle in the middle; drag to 30/70, 50/50, or 70/30, or all the way to close one.
- Drag text, images, and files from one app to the other.

Floating window (v2)

- Any app can shrink into a floating Regular-glass window, like picture-in-picture for all apps.
- Drag by its top handle, pinch to resize, fling to an edge to tuck it away as a small tab.
- Video apps enter picture-in-picture automatically when you go Home.

App lifecycle (what the user feels): apps resume instantly from the switcher. The system freezes background apps after 10 s and restores their last screen after memory pressure, so closing apps manually is never needed.

Keyboard and text input

The keyboard is a Thick-glass panel with swipe typing, a smart strip on top, and first-class Hindi and Hinglish, and users can replace it with any third-party keyboard.

Layout (total 290 pt tall incl. gesture area)

1. **Smart strip (44 pt):** 3 word suggestions; changes to clipboard items, emoji suggestions, or Intent Bar actions depending on context.
2. **Letter rows (3 x 54 pt):** keys are 34 x 46 pt glyph areas with full-cell touch targets; no visible key borders, just subtle glass bumps.
3. **Bottom row (54 pt):** 123, emoji, space bar (wide), language, return.
4. **Below:** globe/mic row in the gesture area.

Features

- Swipe typing, autocorrect with a one-tap undo chip, and a press-and-hold popup for accented letters and numbers on the top row.
- **Space bar trackpad:** long-press space to move the cursor; add a second finger to select text.
- **Clipboard history:** last 20 items for 1 hour, then cleared; sensitive items (passwords, OTPs) are never stored.
- **OTP autofill:** codes from SMS appear in the smart strip.
- **Languages:** English, Hindi (Devanagari), Hinglish (Latin script that suggests in either), plus other Indic scripts; swipe on the space bar to switch.
- Voice typing on-device, with punctuation spoken or automatic.
- One-handed mode: long-press the emoji key and pick left or right; the keyboard shrinks to 80% and docks to that side.

Text selection everywhere: double-tap selects a word, triple-tap selects a paragraph. A glass pill menu floats above the selection with Copy, Paste, Share, Translate and Ask (sends the text to the Intent Bar).

Settings, permissions and privacy

Settings is searchable from anywhere, organized into 10 top sections, and every permission is visible, revocable, and logged.

Settings layout

1. Large title "Settings", with a search field at the **bottom** (floating glass).
2. **Account card** at the top: name, photo, storage used, backup status.
3. Grouped glass lists, each row with a colored symbol, title and current value on the right.
4. Tapping a row pushes a new page sliding in from the right; back swipe from either edge returns.

Top-level sections

1. Network and connections (Wi-Fi, Mobile, Bluetooth, Hotspot, VPN)
2. Notifications and Focus
3. Display and glass (brightness, dark mode, text size, transparency, refresh rate)
4. Home, lock screen and wallpaper
5. Sound and haptics
6. Privacy and security
7. Battery
8. Apps (defaults, permissions per app, storage per app)
9. Accessibility
10. System (updates, backup, language, date and time, developer options, about)

Permission model

Permission	Options offered	Auto-revoke
Location	Precise or approximate; While using, Only this time, Deny	After 30 days unused
Camera, Microphone	While using, Only this time, Deny	After 30 days unused
Photos, Files	Selected items only, All, Deny	After 30 days unused
Contacts, Calendar	Selected, All, Deny	After 30 days unused
Notifications	Allow per channel, Deny	Never
Network access	Allow, Wi-Fi only, Deny (v2)	Never
Background activity	Allowed, Optimized, Restricted	Never

Privacy surfaces

- **Indicators:** a green dot (camera), orange dot (mic), or blue arrow (location) appears next to the Halo whenever one is in use; tap it to see which app.
- **Privacy dashboard:** a 7-day timeline per permission showing which app used it and when.
- **Clipboard alert:** a small toast when any app reads the clipboard.
- **App report:** per app, what it accessed, which domains it contacted (v2), and a one-tap "Revoke all".

Signature features

Ten features that make Sphatik a different kind of phone rather than a reskin. Each plays to what Sphatik really is: real Linux, private by default, built in India. All ten are interactive in the prototype.

#	Feature	Group	Release
1	Desktop mode	Real Linux	v2
2	Terminal and Linux apps	Real Linux	v1 (Terminal), v2 (Linux desktop apps)
3	Halo Shelf	Signature interactions	v1
4	Glass Lens	Signature interactions	v2
5	Living wallpapers	Signature interactions	v1
6	Every screen in your language	Built for India	v1 (Hindi), v2 (all packs)
7	Offline Nearby	Built for India	v2
8	Private Memory	Privacy you can see	v2
9	Tracker firewall	Privacy you can see	v1 (see Privacy system)
10	Longevity promise	Longevity	v1

1. Desktop mode

- Connect a display over USB-C (with a hub that carries video) or cast wirelessly; the compositor switches the external screen to a desktop layout.

- Layout: wallpaper, a floating glass status pill at the top (Sphatik label, date and time, Wi-Fi, battery), a glass dock at the bottom, and every app in a resizable glass window with its icon, title, minimise and close.
- Windows can be dragged, overlapped and focused; the phone screen becomes a trackpad and keyboard, or a Bluetooth keyboard and mouse can be used.
- It is the same compositor with a different layout mode, not a second system; apps adapt through the Medium and Expanded size classes.
- Start: Control Center → Desktop tile, or Settings → Desktop mode.

2. Terminal and Linux apps

- **Terminal** is the 17th stock app, dark, full screen, with the `sphsh` shell, command history, and the Sphatik glass status bar. It lives in the App Library under a new **Developer** category (Terminal, Files, Browser, Notes).
- Runs real tools on the phone: Python, Git, SSH, compilers, and the `sp` developer tool.
- v2 adds Linux desktop apps from a curated repository, running sandboxed like native apps; they are most useful in Desktop mode.
- Terminal apps get no extra privileges: the same sandbox and permissions apply, and there is no `sudo` for apps.

3. Halo Shelf

- Long-press any message, photo, note, file or story, then drag it up; the Halo glows and grows as a drop target.
- Drop it on the Halo to park it. A small black droplet with a count appears: beside the Halo when idle, hanging beneath it when a live activity is running, hidden while the Halo is expanded.
- Tap the droplet for the Shelf panel (Thick glass): up to 8 items, each with **Use** and remove. **Use** drops the item into what you are doing: text goes into the message composer or the open note, photos are sent into the open chat; elsewhere text is copied.
- Items can also be sent to nearby phones through Offline Nearby, or added from Glass Lens.

4. Glass Lens

- Two-finger long-press anywhere (also the Lens tile in Control Center, or Alt+L with a keyboard).
- The screen blurs behind a glass layer and all visible text lifts out as selectable glass chips.
- Actions: **Copy**, **Hindi** (translate the selection), **Read aloud**, **To Shelf**. v2 adds text from images (on-device text recognition) and **Explain** with the on-device model.

- Nothing leaves the phone; the header says so every time.

5. Living wallpapers

- Ribbons and crystals shift with tilt (parallax), and the lock-screen depth crystal moves with them.
- Colour follows the time of day: warm at dawn, clear by day, amber at dusk, deep blue at night.
- The crystal wallpapers glow with battery level; v2 adds weather tint and a charging pulse.
- Motion updates only when the phone moves or the time changes, never as a constant animation, so glass stays cheap to render.

6. Every screen in your language

- **Show the phone in Hindi:** every screen switches instantly and offline, including app names, tabs, settings, widgets and sample content (Settings → Language and translation; deliberately not in Control Center, since most users keep one language).
- **Read aloud** in Hindi or English anywhere, through Glass Lens or the Language page.
- **Hinglish keyboard:** type in Latin letters and get Hindi or English suggestions.
- **Offline language packs** (about 40 MB each): Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada, Malayalam, Punjabi and Odia, downloaded from the Language page.
- The prototype translates a built-in set of phrases; Sphatik uses an on-device translation model for any text.

7. Offline Nearby

- Share files and chat with nearby Sphatik phones with no internet, over Wi-Fi Direct and Bluetooth.
- **Share:** a radar shows nearby phones with names and distance; tap **Send** to send the latest Shelf item or a chosen file.
- **Chat:** local group rooms (for example "College fest volunteers") that work in crowded places with no signal.
- Only phones the user can see nearby can join; messages are end-to-end encrypted and stored only on the phones in the room.
- Start: Control Center → Nearby tile.

8. Private Memory

- Ask the Intent Bar a question such as "What was the Wi-Fi password Aman sent?"; questions are recognised by a question mark or words like what, where, when, who, find.

- On-device AI searches messages, notes, mail and screenshot text, and answers with the exact line, who it came from, the app and when, with the matching words highlighted.
- Every answer ends with "Found on this phone. Nothing was sent anywhere."
- The index is encrypted, can be paused or deleted, and excludes Private Space, password fields and any app the user opts out (see Privacy system).

9. Tracker firewall

Blocks known tracking servers system-wide, including inside Android apps, and produces the weekly "who tried to track you" report. Full behaviour, screens and release plan are in the Privacy system section below; in the prototype it is under Settings → Privacy and security.

10. Longevity promise

- Stated targets, shown in Settings → Battery: updates for this phone until at least 2031, idle memory under 750 MB on 4 GB, 60 fps glass tuned for the Snapdragon 720G.
- **Protect battery:** stop charging at 80% (on by default) and show battery health.
- **Keep this phone fast:** apps unused for 30 days are put to sleep and their caches trimmed.
- The positioning: Sphatik makes old phones feel new, against phone makers who stop updating devices after two years.

Control Center additions: a new row of labelled tiles: Nearby, Lens and Desktop.

Settings additions: Language and translation, Desktop mode, and new pages behind Privacy and security and Battery.

Privacy system

Sphatik treats every app as untrusted until the user says otherwise: nothing is allowed by default, every access is visible, and the user can give an app less than it asks for without breaking it.

Principles

1. **Default deny.** No capability is granted at install, including the internet.
2. **See everything.** Every access and every server an app contacts is logged and shown in plain language.
3. **Finer than yes or no.** Time limits, decoy data and per-server control instead of a single Allow button.
4. **Nothing hidden.** Background use of sensitive capabilities is always visible in the Halo.
5. **Local by default.** The OS collects no data; AI runs on the phone.

6. Honest limits. Where the hardware cannot protect the user, Sphatik says so.

Capabilities and defaults

Capability	Default	Grant options
Internet	Denied	Allow, Wi-Fi only, per-server rules (see Network privacy)
Local network and nearby devices	Denied	While visible, Always
Location	Denied	Precise or approximate; Only this time, While visible, For 1 hour, Decoy
Camera, microphone	Denied	Only this time, While visible, For 1 hour
Motion and environment sensors (accelerometer, gyroscope, barometer)	Denied	While visible; always denied in the background
Contacts, calendar	Denied	Selected entries, All, Decoy
Photos and videos, files	Denied	Selected items (system picker), All, Decoy
Notifications	Denied	Allow per channel
Background activity	Denied	Allowed only with a visible live activity or an explicit grant
Clipboard	No permission exists	Apps receive clipboard contents only at the moment the user pastes into them
Device identifiers (IMEI, serial, SIM serial, MAC)	Never available	None; each app gets a random per-app ID the user can reset

Time-boxed choices are offered on every sensitive prompt: **Only this time**, **For 1 hour**, **Until the screen turns off**, **While visible**.

The permission prompt

- A Thick-glass sheet drawn by the shell; apps cannot draw or imitate it, and it always shows the real app icon and name.
- Shows the one-line reason from the app's manifest, then the grant options, then **Use decoy** and **Deny**.
- Appears only at the moment of use, never at first launch in a batch.

- If the user denies twice, the app cannot ask again until the user opens its permission page.

Decoy data (the app keeps working and learns nothing real)

Capability	What the app receives
Contacts	An empty address book
Photos and files	An empty library or folder
Calendar	An empty calendar
Location	The centre of the user's city, never updated more than hourly

Identifiers

- Each app gets its own random identifier, unrelated across apps; the user can reset it from the app's page.
- Wi-Fi hardware address randomised per network and rotated daily; Bluetooth address randomised; device name on networks is random.
- There is no advertising ID.

Network privacy

- **Per-app internet permission**, enforced by a firewall that knows which app owns each connection (native and Android apps alike).
- **Domain firewall**: each app's page lists every server it contacted; the user can block any of them individually.
- **Tracker blocking** on by default using community blocklists stored and matched on the phone, updated weekly.
- **Encrypted DNS** always on, with a user-chosen resolver.
- **Per-app VPN routing** so selected apps can go through a VPN while others do not.

Privacy Ledger

- Records every access: time, app, capability, foreground or background, and detail (for example the domain contacted or the approximate location returned).
- Kept for 30 days in credential-encrypted storage, hash-chained so entries cannot be quietly edited or removed.
- Never uploaded; the user can export it as a file or delete it.
- Browsable by app ("what did this app do?") or by capability ("who used my microphone?").

Verified behaviour labels: app stores usually show privacy labels written by the developer. Sphatik compares the declared label with what the Ledger actually observed over the first 7 days and shows **Claims** next to **Actually does** on the app's page and in the Sphatik Store. Mismatches are highlighted.

Weekly privacy report: delivered on Sunday evening as a Summary notification. It covers which apps used sensitive capabilities, how many tracker connections were blocked and by which apps, any permissions auto-revoked, and one suggested action ("Instagram used location 40 times this week; switch to approximate?").

No hidden activity

- Camera (green), microphone (orange) and location (blue) dots next to the Halo, drawn by the compositor so apps cannot hide them.
- Any use of these, or of the network, by an app the user is not looking at appears in the Halo as a compact indicator with that app's icon; tapping it opens the Ledger entry.
- **Kill switches** for camera, microphone and location in Control Center cut access for every app at once.

Auto-revoke: permissions for apps unused for 7 days are removed automatically, and time-boxed grants expire on schedule. A monthly summary notification lists what was removed.

Device protection

Feature	Behaviour
Lockdown	Hold the power button and tap Lockdown: disables fingerprint and face unlock, USB data and notification previews until the next PIN entry
Duress PIN	A second PIN set by the user. Entering it opens a decoy profile, or erases Private Space if the user chose that during setup (confirmed twice, irreversible)
Private Space	A separate profile with its own encryption key and PIN; its apps, notifications and files are invisible from the main profile
Auto-reboot	After a user-set idle period (default 18 hours), the phone restarts so all personal data returns to its fully encrypted state
USB data	Off whenever the phone is locked; charging still works

Feature	Behaviour
Lock screen	Notification contents hidden by default; per-app override
Sensitive fields	Password and payment fields are excluded from screenshots, screen recordings and the app switcher snapshot

Android apps

- Pass through the same permissions, firewall, Ledger, decoy data and indicators as native apps.
- Receive random per-app identifiers; the container cannot see Sphatik's files except the shared folder the user controls.
- microG runs as an ordinary unprivileged app inside the container, never with system power.

On-device AI

- Private Memory and the Intent Bar run on the phone. The Memory index is encrypted, can be paused, and excludes Private Space, password fields and any app the user opts out.
- If the user enables a cloud model, Sphatik shows exactly what will be sent before each request, and cloud use appears in the Ledger.

Privacy Center (Settings → Privacy): a summary card (this week's accesses, trackers blocked, apps with background access), then Privacy Ledger, Weekly report, Permissions by type, Network and firewall, Identifiers, Lockdown and Duress PIN, Private Space, and Security limits.

Honest limits (shown under Security limits): privacy against apps and companies can exceed Android and iOS in software. Physical security against a thief with forensic tools cannot: the Redmi Note 9 Pro's bootloader stays unlocked and its security chip is not usable by a custom OS, so encryption keys are protected in software only. Tracker lists are never complete, and no system can stop an app misusing data the user chooses to give it.

Release plan

v1	v2
Default-deny capabilities, time-boxed grants, per-app internet permission, random identifiers, paste-only clipboard, indicators and kill switches, 30-day Privacy Ledger, auto-revoke, Lockdown, USB data lock, auto-reboot	Decoy data, domain firewall with per-server blocking, verified behaviour labels, weekly report, Duress PIN, Private Space, per-app VPN routing

Core apps

Sphatik ships 16 stock apps. Each follows the standard app structure (large title, content under glass, floating glass tab bar), and all 16 are designed and interactive in the prototype.

App	Structure	Key screens and behaviour
Phone	Favorites, Recents, Contacts, Keypad	Recents filter (All/Missed) with colour-coded call types; A to Z contacts with a letter index; keypad with letter hints. In-call screen with Mute, Keypad, Speaker, Add call, Video, Hold and End; the call collapses into a green Halo when you leave the app
Messages	Single list	Search, pinned contacts, unread dots; conversations with a glass composer, typing indicator and spring-in bubbles
Browser	Page + floating address bar	Start page with shortcuts, reading list and top stories; bottom address bar accepts searches; tab overview grid
Settings	Single list	Search, account card, wallpaper picker, the ten top-level sections
Calendar	Month view + agenda	Month grid with today and event dots; tap a day for its agenda; month navigation; add button
Clock	Alarms, World, Timer, Stopwatch	Alarm cards with toggles; world clocks with live mini dials; timer ring that keeps counting in the Halo; stopwatch with laps

App	Structure	Key screens and behaviour
Weather	Full-screen sky	Hourly strip, 8-day forecast with range bars, tiles for air quality, UV, feels like, humidity, wind and sunset
Maps	Full-screen map	Search, layers, location dot, nearby chips and recents sheet; route card with ETA and travel modes; turn-by-turn banner
Gallery	Photos, Albums, Search	Day, month and year views; albums and media types; people and places; full-screen viewer with favourite, edit and delete
Camera	Viewfinder	Night, Photo, Video and Portrait modes; zoom; tap to focus; shutter flash; recording timer; privacy dot when the real camera is on
Calculator	Single view	Order of operations, running history line, Indian digit grouping
Notes	Notes, Folders	Folder filters, pinned cards, editor with tappable checklists and a floating toolbar
Files	Recents, Browse, Storage	File-type icons, categories and locations, storage ring by category with a cleanup action
Store	Discover, Apps, Updates, Search	App of the day, editors' picks, top charts, Update all with progress indicators
Music	Library, Search, Now Playing	Recently added grid, genre tiles, full-screen player over blurred album colours, mini player; stays in sync with the Halo and Control Center
Mail	Inbox	Mailbox chips, unread markers, read view with Reply and Forward, compose screen with a Sent banner

Shared component library every app uses: glass title bar, floating glass tab bar, search field, segmented control, toggle, list row and grouped card, avatar, floating action button, pushed view with back navigation, bottom sheet, menu, toast and banner, slider, text field and empty state.

Accessibility and performance budgets

Glass is only beautiful if everyone can read it and it never stutters, so these limits are hard requirements, not goals.

Accessibility (v1)

- Screen reader with gestures (swipe to move focus, double-tap to activate); every component in the shared library has labels and roles built in.
- Text size 80% to 310%; layouts reflow instead of truncating.
- Reduce Transparency, Reduce Motion, Increase Contrast, Bold Text, and color filters for color blindness.
- Minimum 4.5:1 text contrast on all glass (enforced by the Legibility guard).
- 3-button navigation and a floating Assistive button as alternatives to gestures.
- Captions for all system audio and video; mono audio and balance control.

Performance budgets (measured on the target device, Redmi Note 9 Pro)

Metric	Budget
Frame rate during any system animation	60 fps locked, zero dropped frames in shell transitions
Touch to first pixel response	Under 20 ms
Cold boot to lock screen	Under 15 s
Unlock to usable home	Under 300 ms
First-party app cold launch	Under 500 ms
Glass compositing cost per frame	Under 3 ms GPU at 1080 x 2400
Idle system RAM	750 MB or less
Standby battery drain	Under 1% per hour

Graceful degradation order when a frame budget is at risk: 1) drop caustics, 2) drop refraction and dispersion, 3) halve blur radius, 4) use a cached blurred snapshot instead of live blur, 5) solid tint.

Target device: Redmi Note 9 Pro

Sphatik v1 targets the Redmi Note 9 Pro (miotoll family), running on a mainline Linux kernel, locked at a smooth 60 fps on its 60 Hz LCD. This is realistic because the community postmarketOS port already runs the core hardware ([source](#)).

Hardware profile (development unit: India model, codename curtana, 4 GB RAM, 64 GB storage)

Part	Spec	What it means for Sphatik
SoC	Snapdragon 720G: 2x Cortex-A76 + 6x Cortex-A55	UI thread and compositor pinned to the 2 big cores; background work on the 6 small cores
GPU	Adreno 618	Open Mesa drivers (Freedreno GL, Turnip Vulkan); glass renders via Vulkan
Display	6.67" 1080 x 2400 IPS LCD, 60 Hz, center punch-hole	16.6 ms frame budget; the Halo sits perfectly around the centered camera
RAM	4 to 8 GB depending on variant	Design for the 4 GB model so every variant is smooth
Storage	UFS 2.1	Fast enough for sub-400 ms app launches
Battery	About 5000 mAh	LCD gains nothing from dark pixels, so Always-on display costs real power
Unlock	Side-mounted fingerprint in power button	Not working on mainline yet

Mainline Linux status on miatoll (community port, testing tier)

Works	Broken or missing	Needs a fix
Wi-Fi, calls, mobile data, audio, GPU, sensors	Fingerprint, NFC, hardware video decode	Camera (a community kernel patch exists)

Source: [asidko/redmi-note-9s-postmarketos](https://github.com/asidko/redmi-note-9s-postmarketos). The port also notes 3 boot gotchas: the Plymouth boot splash hangs, the boot partition must be shrunk to fit the small cache partition, and the battery gauge reports garbage for the first few seconds.

Design changes for this hardware

- **60 fps, locked.** All springs are tuned at 60 Hz. The 120 fps path stays in the code for future high-refresh devices.
- **Glass tier "Balanced"** by default: blur computed at quarter resolution and upscaled (looks the same behind 24+ px blur), at most 2 live-blur layers on screen, any extra glass panels use a cached blurred snapshot, caustics off during scrolling.

- **Always-on display off by default**, replaced by "lift to glance": raising the phone wakes the lock screen for 5 s at low brightness.
- **Unlock in v1** with PIN, pattern, or password. Fingerprint joins when the mainline driver works; face unlock comes later with the on-device AI.
- **Video**: software decoding on the big cores, capped at 1080p30 H.264/VP9/AV1, until hardware decode lands upstream.
- **Battery service** ignores fuel-gauge readings for the first 10 s after boot, so the phone never shuts down on a false "0%".
- **Boot splash**: the compositor draws the SKY boot logo (the SKY wordmark centred on screen, with "SPHATIK OS" underneath), so it does not use Plymouth.
- **On-device AI**: a small model (about 1 billion parameters, 4-bit, under 800 MB in RAM) only, loaded on demand and unloaded when idle so it never costs RAM in the background. Larger models are cloud-only, with opt-in.

Memory budget (4 GB variant)

Component	Budget
Kernel + system services	350 MB
Compositor + shell (incl. glass buffers)	250 MB
Keyboard, Intent Bar, background services	150 MB
Total system at idle	750 MB or less
Left for apps	About 3 GB

Storage budget (64 GB unit)

Component	Budget
Sphatik system image (A/B slots, 2 copies)	2 x 3 GB
On-device AI model	800 MB
Fonts, wallpapers, system sounds	300 MB
Reserved for updates and cache	2 GB
Left for user apps and files	About 50 GB

With 4 GB of RAM, a compressed swap in RAM (zram, 2 GB) is on by default so more apps stay alive in the background.

Bring-up plan on your phone

1. Back up everything; unlocking the bootloader wipes the phone. Use a second phone as your daily driver during development.
2. Unlock the bootloader with Xiaomi's official Mi Unlock tool (it enforces a waiting period).
3. Flash the postmarketOS miatoll image with Phosh to confirm calls, Wi-Fi and GPU work on your unit.
4. Replace Phosh with the Sphatik compositor and shell, running on DRM/KMS and Vulkan, on top of the postmarketOS base.
5. Later, replace the base with Sphatik's own init and system image.

The modem, persist and other firmware partitions survive this process, so returning to MIUI stays possible.

Prototype status

The interactive shell prototype is live at [Sphatik OS prototype](#). It runs in any browser and is best judged on the Redmi Note 9 Pro itself.

Built

- **System:** SKY boot, lock screen with complications and customisation, home with widgets and edit mode (drag to rearrange, remove, add widgets), App Library with categories, Notification Center, Control Center, Intent Bar, app switcher and banners.
- **Halo live activities:** call, timer and media.
- **Apps:** all 16 stock apps plus Terminal, the ten signature features, and the Privacy Center with the tracker firewall, Privacy Ledger and weekly report.
- **Look:** lens-light glass on every surface, four wallpapers, jewel-tone app icons with a live Calendar date and Clock.

Not built yet: keyboard, Focus modes, split screen and floating windows, notification summaries, real refraction (belongs to the native compositor), and anything that touches real phone data.

Next: Phase 2, the Rust compositor that renders this same UI on the device.

Open questions

These decisions shape the next round of design and should be settled before mockups start.

- ☐ Name decided: Sphatik OS. Still to check: trademark search on IP India, plus domain and GitHub org availability.

- ☐ App model: Sphatik-native apps only in v1, or Android app compatibility from day one?
- ☐ Default home: free placement with auto-arrange off, or on?
- ☐ Intent Bar trigger: swipe down on home only, or also a dedicated button or side-button press?
- ☐ On-device AI model: which size fits the target phone's RAM alongside the OS budget?
- ☐ Target device decided: Redmi Note 9 Pro. India model (curtana), 4 GB RAM, 64 GB storage.
- ☐ Font build route: draw Sphatik Sans fully from scratch, or fork an OFL font (Inter or Manrope) and redraw its signature details?
- ☐ Store policy: who can publish, and how are apps reviewed?